TN4 - asd

Spanish II (Saigon South International School)

Thiet ke he thong va vi mach tich hop_ Nhom 11

Bắt đầu vào lúc	Thursday, 18 May 2023, 10:02 PM
Trạng thái	Đã xong
Kết thúc lúc	Thursday, 18 May 2023, 10:25 PM
Thời gian thực hiện	22 phút 15 giây
Điểm	**21,50** trên 25,00 (**86**%)

Câu hỏi 1

Đúng

Đạt điểm 1,00 trên 1,00

1. If A= 4`b011 and B= 4b`0011, then the result of AB will be**

Select one:

a. 9

b. Invalid expression

c. 6

d. 27 ✔

Your answer is correct.

Câu hỏi 2

Sai

Đạt điểm 0,00 trên 1,00

Initial value of a=1 and b=2, then what will be final value if

always @ (posedge clock)

a=b;

always @ (posedge clock)

b=a;

Select one:

- a. None of the above ✖
- b. a= 2, b=1
- c. a= 1, b=2
- d. Both a and b will have same value either 0 or 1

Your answer is incorrect.

Câu hỏi 3

Đúng

Đạt điểm 1,00 trên 1,00

In verilog `h1234 is a

Select one:

- a. 16 bit hexadecimal number
- b. 32 bit hexadecimal number ✔
- c. It is invalid notation
- d. 4 bit hexadecimal number

Your answer is correct.

Câu hỏi 4

Đúng

Đạt điểm 1,00 trên 1,00

What will the following code segment generate on synthesis, assuming that the four variables data0, data1, data2 and data3 map into four latches/ flip-flops?

```
always @(posedge clock)
    begin
    data3 = din;
    data2 = data3;
    data1 = data2;
    data0 = data1;
    end
```

Select one:

- a. A 4-bit parallel-in parallel-out register
- b. A 4-bit shift register
- c. Four D flip-flops all fed with the data “din”
- d. None of these

Your answer is correct.

Câu hỏi 5

Đúng

Đạt điểm 1,00 trên 1,00

Consider the following Verilog code segment:

wire [5:0] A, B;

wire C;

assign C=^A;

If the values of A and B are 5'b10011 and 5'b01110 respectively, what will be the value of **{A[3:1], 2{C}, B[2:0]}**?

Select one:

- a. 00111110 ✔
- b. 00100110
- c. None of these
- d. 01111110

Your answer is correct.

Câu hỏi 6

Đúng

Đạt điểm 1,00 trên 1,00

In the given code snippet, statement 2 will executed at

initial

begin

#5 x= 1'b0; // statement 1

15 y= 1b'1; //statement 2

end

Select one:

- a. 5
- b. 20 ✔
- c. 15
- d. Current simulation time

Your answer is correct.

Câu hỏi 7

Đúng

Đạt điểm 1,00 trên 1,00

Which of the following is true about the following code segment

integer x, y

```
;
initial
        begin
        x = 15;
        y = 10;
        end
initial
        repeat (x) $display (“x=%d”, x);

initial
        while (y < 12)
            begin
                y = y + 1;
                x = x − 1;
            end
```

Select one:

- a. The simulation will always display 15 as the value of ‘x’
- b. It cannot be determined exactly how many times the value of ‘x’ will be printed
- c. The simulation will print the current value of ‘x’ 13-times
- d. The simulation will print the current value of ‘x’ 15-times ✔

Your answer is correct.

Câu hỏi 8

Đúng

Đạt điểm 1,00 trên 1,00

Which of the following statements are true?

(multiple choices)

Select one or more:

- [x] a. A variable index used at the RHS of an “assign” statement generates a multiplexer whereas usage of it at the LHS results in a decoder ✔
- [x] b. The “assign” statement can be used to implement both combinational as well as sequential circuits ✔
- [ ] c. The use of conditional operator in an “assign” statement always results in a multiplexer
- [ ] d. All of the above

Your answer is correct.

Câu hỏi 9

Sai

Đạt điểm 0,00 trên 1,00

What does the following code segment implement?

```
assign d = ~(c | b);
assign c = ~(a | d);
```

Select one:

- ( ) a. A 2-bit comparator
- ( ) b. A 2-bit shift-register
- ( ) c. A 1-bit latch
- (•) d. Two NOR gates connected in cascade ✖

Your answer is incorrect.

Câu hỏi 10

Đúng

Đạt điểm 1,00 trên 1,00

What is the purpose of the “initial” procedural block in Verilog test benches?

Select one:

- a. If there are multiple “initial” blocks in a module, they are executed one by one in sequence
- b. It is used to specify a procedural block that is executed only once ✔
- c. It can be used to specify a procedural block for synthesis
- d. Every time an “always” block is executed, variables are initialized as specified in the “initial” block

Your answer is correct.

Câu hỏi 11

Đúng

Đạt điểm 1,00 trên 1,00

Which is not a correct method of specifying time scale in verilog?

Select one:

- a. 100ns/110ps ✔
- b. 10ns/1ps
- c. 100ns/100ps
- d. 1ns/1ps

Your answer is correct.

Câu hỏi 12

Đúng

Đạt điểm 1,00 trên 1,00

What is meant by synthesis?

Select one:

a. To verify whether the specification is correct.

b. To generate a circuit from a given specification.

c. None of these.

d. To verify the correct operation of a circuit.

Your answer is correct.

Câu hỏi 13

Đúng

Đạt điểm 1,00 trên 1,00

If A, B, C and D are reg, reg, integer and wire variables respectively, each of size [7:0], which of the following is/are allowed inside a procedural block? **(Multiple choice)**

Select one or more:

a. D = C + 1;

b. D = A + B;

c. B[3:0] = D[4:1] + 1;

d. C = A + D;

Your answer is correct.

Câu hỏi 14

Đúng

Đạt điểm 1,00 trên 1,00

What does the construct "#5" indicate in simulation?

Select one:

- a. It schedules the execution of the next statement at time 5
- b. It specifies a delay of 5 time units before executing the next statement ✔
- c. It pauses execution of the statements that follow after time 5
- d. It specifies that the unit of delay is 5 nanoseconds

Your answer is correct.

Câu hỏi 15

Đúng

Đạt điểm 1,00 trên 1,00

If "clk" and "clear" are two inputs of a counter module, which of the following event expressions must be used if we want to implement asynchronous clear (assuming "clear" is active low, active high edge of "clk" signal is used for counting, and single "always" block is used for the implementation)?

Select one:

- a. always @(posedge clk)
- b. None of these
- c. always @(negedge clear)
- d. always @(posedge clk or negedge clear) ✔

Your answer is correct.

Câu hỏi 16

Đúng

Đạt điểm 1,00 trên 1,00

Which of the following is true for the “repeat” loop?

Select one:

- a. It can be used to iterate a block indefinitely
- b. It can be used to repeat execution of the block exactly two times
- c. It can be used to iterate a block until a specified condition is true
- d. None of these ✔

Your answer is correct.

Câu hỏi 17

Đúng

Đạt điểm 1,00 trên 1,00

In a pure combinational circuit is it necessary to mention all the inputs in sensitivity list?

Select one:

- a. None of these
- b. Yes ✔
- c. No
- d. It depends on the coding style

Your answer is correct.

Câu hỏi 18

Đúng

Đạt điểm 1,00 trên 1,00

For the following code segment, the final value of variable “d” will be … integer a, b, c, d; initial begin a = 25; b = 12; c = 5; d = 17; a = b + c; b = a − 15; c = a + d; d = c + d; end

Select one:

- a. 53
- b. 40
- c. 51 ✔
- d. 58

Your answer is correct.

Câu hỏi 19

Đúng

Đạt điểm 1,00 trên 1,00

Which of the following is true for the following module?

```
module mydesign (a,b);
input [1:0] b;
output reg a;
always @(b)
  begin
    if (b==2'b00) a = 1'b0;
    else if (b==2'b11) a = 1'b0;
    else a = 1'b1;
  end
end module
```

Select one:

- a. The synthesis tool will give an error
- b. A combinational circuit implementing an AND function will be generated
- c. A combinational circuit implementing a XOR function will be generated. ✔
- d. A latch will be generated for the output "a"

Your answer is correct.

Câu hỏi 20

Đúng

Đạt điểm 1,00 trên 1,00

If a net has no driver, it gets the value

Select one:

- a. 0
- b. Z ✔
- c. U
- d. X

Your answer is correct.

Câu hỏi 21

Đúng một phần

Đạt điểm 0,50 trên 1,00

Which of the following statements is false for Verilog modules?

(multiple choices)

Select one or more:

- [x] a. If a module X is instantiated 4 times within another module, 4 copies of X are created. ✘
- [x] b. When a module X is called multiple numbers of times from some other module, only one copy of module X is included in the hardware after synthesis. ✔
- [ ] c. A module cannot contain definitions of other modules.
- [ ] d. More than one module cannot be instantiated within another module.

Your answer is partially correct.

Bạn đã chọn đúng 1.

Câu hỏi 22

Đúng

Đạt điểm 1,00 trên 1,00

Default value of reg is

Select one:

- a. Z
- b. X
- c. 0
- d. U

Your answer is correct.

Câu hỏi 23

Đúng

Đạt điểm 1,00 trên 1,00

Consider the following Verilog module.

module guess (data, cond, result);

input [7:0] data;

input [1:0] cond;

output reg result;

always @(data)

begin

if (cond == 2'b00)

result = |data;

else result = ~^data;

end

endmodule

Which of the following are true when the module is synthesized?

(multiple choices)

Select one or more:

- [ ] a. A sequential circuit with a storage element for result will be generated.
- [ ] b. None of the above.
- [x] c. The synthesize system will generate a wire for result. ✔
- [x] d. A combinational circuit will be generated. ✔

Your answer is correct.

Câu hỏi 24

Không trả lời

Đạt điểm 1,00

Which of the following is true for the following code block?

```
output reg clk1, clk2;
initial
begin
clk1 = 1'b0;
clk2 = 1'b1;
forever clk1 = !clk1;
repeat (5) #5 clk2 = ~clk2;
#75 $finish;
end
```

(multiple choices)

Select one or more:

- a. The variable “clk2” will never change its initial state
- b. The variable “clk1” will change its state indefinitely until terminates at time 75 units
- c. The block will never terminate
- d. The state of “clk2” variable will change 50 times

Your answer is incorrect.

Câu hỏi 25

Đúng

Đạt điểm 1,00 trên 1,00

Which of the following statements is/are true?

Select one or more:

- [ ] a. None of these
- [x] b. The "assign" statement implements continuous assignment between the expression specified on the right-hand side and a "net" type variable specified on the left-hand side ✔
- [x] c. The "assign" statement can be used to model a latch, which is a sequential circuit ✔
- [ ] d. The "assign" statement implements continuous assignment between the expression specified on the right-hand side and a "reg" type variable specified on the left-hand side

Your answer is correct.

◄ Quiz 2 - Chap3

Chuyển tới...

Chapter 04 Combination Logic Circuit ►